F78AA - Actuarial Financial Mathematics, December 2010, Solutions

1. (a) $(1+i)^8 = 1.16 \;\; \Rightarrow \mathrm{i} = (1.16)^{(1/8)} - 1 = 1.87257$ ①

   (b) $(1+i)^{1/0.9} = 1.013^3 \;\Rightarrow\; i = 1.013^{3\times 0.9} - 1 = 3.548903\%$ ①

   (c)

$$\mathrm{PV}(31Dec11) = 150\times 1.05^{3/12}\times 1.008^5 + 120\times 1.008^2 = 158.01243 + 121.92768 = 279.94$$

③

$$\mathrm{PV}(1Jan11) = 279.94 \times 1.05^{-7/12} \times 1.008^{-5} = 261.46$$

①

[Total (Question 1) 6 marks]

2. ‘

(a) $X \times a_{\overline{25}|}^{i=0.05} = £100,000,$
$v^{25} = 1.05^{-25} = 0.295303, \Rightarrow \mathrm{a}_{\overline{25}|}^{i=0.05} = (1 - v^{25})/0.05 = 14.093945$
$\Rightarrow\ X = £7,095.25$ ③

(b) **The Mathematics of the Loan Schedule**

Time (yrs)	**Repayment**	**Interest Content**	**Capital Content**	**Capital Outstanding**
0	-	-	-	$L = X.a_n$
1	X	$X.i.a_n$ $= X(1 - v^n)$	Xv^n	$L - X.v^n = X(a_n - v^n)$ $= Xa_{n-1}$
2	X	$X.i.a_{n-1}$ $= X(1 - v^{n-1})$	Xv^{n-1}	$X(a_{n-1} - v^{n-1})$ $= Xa_{n-2}$
⋮				
t	X	$X.i.a_{n+1-t}$ $= X(1 - v^{n+1-t})$	Xv^{n+1-t}	$X(a_{n+1-t} - v^{n+1-t})$ $= Xa_{n-t}$

④

(c) Capital o/s after 8th Payment $= X \times a_{\overline{17}|}^{i=0.05} = £7,095.25 \times 11.274066 = £79,992.32,$
$\Rightarrow$ Capital o/s at time 12 = £79,992.27 $\times 1.05^4 = £97,231.16,$
$\Rightarrow$ This is now to repaid over 13 years. $\mathrm{a}_{\overline{13}|}^{i=0.05} = 9.393573,$
$\Rightarrow$ New Instalment = £97,231.10 / 9.393573 = £10,350.82

④

(d) Capital o/s after 15th Payment = Capital o/s after 3rd payment under the new 13 year loan Schedule
$\Rightarrow$ Capital o/s after 3rd payment = X $\times a_{\overline{10}|}^{i=0.05} = £10,350.82 \times 7.721735 = £79,926.27$
$\Rightarrow$ Max Incremental Loan = £200,000 - £79,926.27 = £120,073.73 ③ **[Total (Question 2) 14 marks]**

3. (a)

$$\frac{163}{169} \times \frac{105}{126} \times \frac{5200}{5105} \times \frac{170}{100} = 0.9644970 \times 0.8333333 \times 1.0186092 \times 1.7 = 1.391798$$

③

$$1.391798^{1/5} - 1 = 0.068354$$

$\Longrightarrow$ TWRR = 6.835 %. ①

[Total 4 marks]

(b) EoV for $v = 1.0177^{-1} = 0.9826078$:

$$169 - 37v^{0.75} + 5000v^{1.6} - 5100v^3 - 170v^5 = -0.149139 \approx 0$$

③

[Total 3 marks]

(c) The relatively low return on the large investment of 5000.00 means that the overall MWRR is rather low, while TWRR depends equally on all payments independently of the size of the payment. ② **[Total 2 marks]**

[Total (Question 3) 9 marks]

4. (a) An index-linked gilt is a government bond that pays a regular coupon. Coupons and the redemption payment are adjusted according to an inflation index. ①

Therefore, an index-linked gilt gives some protection against the risk of high inflation rates. These gilts are therefore used by investors who want to obtain a certain real yield rather than a monetary yield. ①

[Total 2 marks]

(b) payment on 15 Jan 2013: ②

$$\frac{3}{2}\frac{Q(\text{Oct 2012})}{Q(\text{April 2010})} = \frac{3}{2}\frac{Q(\text{April 2010})}{Q(\text{April 2010})}(1+q)^{2.5} = \frac{3}{2}(1.019)^{2.5} = 1.572269$$

payment on 15 July 2035: ②

$$(100+\frac{3}{2})\frac{Q(\text{April 2035})}{Q(\text{April 2010})} = 101.50\frac{Q(\text{April 2010})}{Q(\text{April 2010})}(1+q)^{25} = 101.50(1.019)^{25} = 162.4878$$

(Marking: 1 mark is given for the right formula, no redemption = -0.5 marks, wrong exponent = -0.5 marks) **[Total 4 marks]**

(c) The investor should invest into a bond which is not index-linked. ①

[Total 1 marks]

[Total (Question 4) 7 marks]

5. (a) The investor obtains a yield of 1.7% per quarter-year. We first check whether or not CGT is payable.

$$\frac{d(1-t_1)}{R} = \frac{2(1-0.4)}{115} = 0.01043478 < 0.017$$

Therefore, CGT is payable. ②

... and the EoV becomes:

$$\begin{aligned} P &= \frac{8}{4}(1-0.4)a_{\overline{60}|}^{0.017} + (115 - 0.18(115-P))(1.017)^{-60} \\ &= 1.2\frac{1-1.017^{-60}}{0.017} + (115 - 0.18\times 115)(1.017)^{-60} + 0.18P(1.017)^{-60} \\ (1-0.18(1.017)^{-60})P &= 1.2\frac{1-1.017^{-60}}{0.017} + (115 - 0.18\times 115)(1.017)^{-60} \\ P &= \frac{1.2\times 37.42944 + 94.30\times 0.3636996}{0.934534} \\ &= \frac{44.91533 + 34.29687}{0.934534} = \frac{79.2122}{0.934534} = 84.76118 \end{aligned}$$

③

[Total 5 marks]

(b) The new investor has to pay CGT as he buys the bond for 84.76118 which is less than the redemption payment of 115. ①

The EoV is therefore

$$\begin{aligned} 84.76118 &= \frac{8}{4}(1-t)a_{\overline{60}|}^{0.02} + (115 - 0.18(115-P))(1.02)^{-60} \\ &= 2(1-t)\times 34.76089 + 109.5570\times 0.3047823 \\ &= (1-t)\times 69.52178 + 33.39103 \end{aligned}$$

②

$$t = 1 - \frac{51.37015}{69.52178} = 1 - 0.7389073 = 0.2610927$$

The investor therefore pays about 26.1 % income tax. ①

[Total 4 marks]

[Total (Question 5) 9 marks]

6. (a) $a^{i}_{\overline{n}|}$ represents the present value at time 0, of an annuity of 1 pa payable in arrears discounted at an effective rate of interest of i pa,
$a^{i}_{\overline{n}|} = v + v^2 + v^3 + ... + v^n$ where $v = \frac{1}{(1+i)}$
$v \times a^{i}_{\overline{n}|} = v^2 + v^3 + ... + v^n + v^{n+1}$, subtracting the 2nd equation from the 1st,
$(1 - v) \times a^{i}_{\overline{n}|} = v - v^{n+1} = v \times (1 - v^n)$
as $\frac{v}{(1-v)} = i$,then
$a^{i}_{\overline{n}|} = \frac{1-v^n}{i}$, as required, Q.E.D.

As we have calculated the present value of the annuity at time zero, we can treat this as one payment of $a^{i}_{\overline{n}|}$ at time 0. Therefore, the accumulation $s^{i}_{\overline{n}|}$ of $a^{i}_{\overline{n}|}$ at time n is given by

$$s^{i}_{\overline{n}|} = (1 + i)^n a^{i}_{\overline{n}|}$$

④

(b) $a^{i}_{\overline{n,even}|}$ represents the present value at time 0, of an annuity of 1 pa payable only for even years in arrears discounted at an effective rate of interest of i pa,
$a^{i}_{\overline{n,even}|} = v^2 + v^4 + v^6 + ... + v^n$ where $v = \frac{1}{(1+i)}$
$v^2 \times a^{i}_{\overline{n,evem}|} = v^4 + v^6 + ... + v^n + v^{n+2}$, subtracting the 2nd equation from the 1st,
$(1 - v^2) \times a^{i}_{\overline{n,even}|} = v^2 - v^{n+2} = v^2 \times (1 - v^n)$
$a^{i}_{\overline{n,even}|} = \frac{v^2}{(1-v^2)} \times 1 - v^n$, but $\frac{v^2}{(1-v^2)} = \frac{1}{i(i+2)}$
$a^{i}_{\overline{n,even}|} = \frac{1}{(2+i)} \times a^{i}_{\overline{n}|}$, as required, Q.E.D.

③

(c) By noting that an annuity that pays only 1 in odd years is given by
$a^{i}_{\overline{n,odd}|} = a^{i}_{\overline{n}|} - a^{i}_{\overline{n,even}|}$, and that $1 - \frac{1}{2+i} = \frac{1+i}{2+i}$
$a^{i}_{\overline{n,odd}|} = \frac{(1+i)}{(2+i)} \times a^{i}_{\overline{n}|}$, or simply by general reasoning that $a^{i}_{\overline{n,odd}|} = (1 + i) \times a^{i}_{\overline{n,even}|}$
Since we have the present value of this annuity at time 0, we can obtain by the same general reasoning as above that the value of the accumulation at time n is given by
$s^{i}_{\overline{n,odd}|} = (1 + i)^n \times \frac{(1+i)}{(2+i)} \times a^{i}_{\overline{n}|}$,
$s^{i}_{\overline{n,odd}|} = \frac{(1+i)}{(2+i)} \times s^{i}_{\overline{n}|}$, as required, Q.E.D.

②

[Total (Question 6) 9 marks]

7. (a) Payback under Lender A : - $(100 + 12.50) \times (1 + (.0075 * 7)) = £118.41$
Payback under Lender B : - $100 * (1 + .0025)^7 + 15 = £116.76$
Decision : - lender B as only £116.76 is required to be paid back ③

(b) The borrower is charged 16.76% for 7 days under Lender B's terms
Therefore the annualised borrowing rate in percentage terms is given by:=
$= ((1 + .1676)^{(365/7)} - 1) \times 100 = 322,679\%$

[Total (Question 7) 6 marks]